Sujal Lohar

23bce334@nirmauni.ac.in — +91 76653 51906 — Ahmedabad, India
LinkedIn — GitHub

Research Profile

Passionate researcher in **Federated Learning, XAI, Quantum Machine Learning, and Healthcare Informatics**. Authored 12+ research papers at the intersection of privacy-preserving AI, quantum-enhanced learning, and explainable systems. Seeking to contribute to cutting-edge research in academia or industry.

Research Interests

- Federated Learning & Privacy-Preserving AI
- Explainable AI (XAI) for Healthcare & Finance
- Quantum Machine Learning & Quantum–Classical Hybrid Models
- 5G-enabled Distributed AI Systems
- Secure, Scalable AI for Real-World Applications

Research Work

1. **X-FedLSTM: An Explainable Federated LSTM Framework for Privacy-Preserving Heart Disease Risk Prediction**
2. **Federated Transformer-Based Learning for Predicting ICU Events with Multi-hop XAI Explanation Models**
3. **X-FED: An Explainable Federated Hybrid IDS for Smart SOC**
4. **FL-QuANT-XAI: A Federated Quantum–Hybrid Framework for Explainable Disease Prediction in Distributed Healthcare**
5. **MitigAI: Novel Frameworks for Identification and Neutralization**
6. **Q-FedX: An Explainable Quantum Federated Learning Approach for Credit Scoring and Fraud Detection**
7. **QuExplain: Explainable AI using Quantum Kernel Methods for Medical Diagnosis**
8. **S-HeartSecure: A Secure, Smart, and Scalable Framework for Heart Disease Risk Prediction using Federated Learning and Blockchain**
9. **SecurePay: A Scalable Big Data Framework for Detecting Credit Card Fraud**
10. **XAI, FL and 5G Integration in Healthcare 5.0**
11. **X-AnalogML: Explainable ML Models for Analog Signal Distortion Mitigation**

12. **Brain Drain Analysis**

    Analyzed causes and impacts of skilled migration in India. Explored policy and opportunity-based solutions to mitigate brain drain.

Technical Skills

- **Languages:** Python, C, C++, Java, Shell Script
- **Frameworks/Libraries:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, React.js, Next.js, Express.js
- **Tools:** Git, Docker, AWS, OCI, Latex, Jupyter, VS Code
- **Concepts:** Federated Learning, XAI, Quantum ML, Blockchain, 5G Networks, OOP, DSA, OS, CN
- **Soft Skills:** Research Writing, Team Leadership, Collaboration, Self-Learning

Professional Experience

T.I.M.E. Pvt. Ltd. — Automation Engineer Intern

- Automated student notifications using Python, Docker, WhatsApp/Twilio APIs, improving communication efficiency by 30%. - Developed a dynamic web platform with React and MongoDB for student engagement.

AIESEC — Junior Manager, Outgoing Global Talent

- Screened 30+ internship applications, shortlisted top 15% candidates. - Managed end-to-end exchange processes for interns across 4 countries.

The Akshaya Patra Foundation — Web Development Intern

- Built a sensory automation website for 2 kitchens, improving reporting efficiency by 30%. - Streamlined daily reporting for 20+ staff, saving 10+ hours weekly.

Projects

Heart Disease Prediction Model

- ML model with 85%+ accuracy on 10k+ entries using Scikit-learn, Pandas, NumPy. - Implemented logistic regression, decision trees, KNN, and evaluated using precision, recall, F1-score.

Music Player Web App

- Responsive music streaming site using Apple Music API, HTML, CSS, JavaScript. - Real-time search, playback, and clean UI/UX for desktop and mobile.

Net Worth Tracker

- Java-based OOP tool for asset/liability tracking with real-time net worth calculation. - Modular design, file I/O for persistence, scalable for personal finance.

Education

B.Tech in Computer Science & Engineering, Minor in Cybersecurity
Nirma University, Ahmedabad *Aug 2023 – Present*
CGPA: 7.76/10.0

Awards & Honors

- **HPAIR Harvard Conference 2025** – Delegate, Japan (Acceptance rate ¡ 10%)
- **Global Goals Summit 2025** – Top 100 Finalist (Asia-Pacific)
- **Ministry of External Affairs Scholarship** – Awarded yearly for academic excellence
- **Rajasthan Government Honoree 2024** – Recognized as scholar student, awarded tablet
- **Jawahar Navodaya Vidyalaya** – Top 1% in national entrance exam
- **Google Developer Group (GDG) Nirma University** – Active member, Google Cloud labs

Languages

- **English:** Advanced
- **Hindi:** Native
- **Gujarati:** Proficient

Volunteering

- **Colgate & MN8 "Bright Smile Bright Future"** – Distributed toothbrush kits, educated children on oral hygiene (Jan–Mar 2025)
- **Community Tech Awareness** – Conducted digital literacy and tech access programs in rural areas